

KUSHAGRA GUPTA

Curriculum Vitae — AI/ML Developer | Data Analyst | Full-Stack Engineer

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PROFESSIONAL PROFILE

Kushagra Gupta is a B.Tech Computer Science (Data Science) student at the Babu Banarasi Das Institute of Technology and Management (BBDITM), Lucknow, with a demonstrated ability to design, build, and deploy AI-powered, data-driven applications end-to-end. His work spans large language model integration, predictive analytics, computer vision, embedded systems, and full-stack web development, consistently favoring production-ready, deployed solutions over theoretical exercises.

Across six completed Forage job simulations with JPMorgan Chase & Co., HPE, Tata, Quantum, and Deloitte Australia, Kushagra has applied quantitative research methods, software engineering practices, generative-AI-powered analytics, cybersecurity analysis, retail data analytics, and business intelligence dashboarding to realistic corporate scenarios. He complements this practical experience with eleven professional certifications, including three from AWS Training & Certification covering generative AI, machine learning engineering, and prompt engineering.

Kushagra is currently focused on deepening his expertise in agentic AI systems, retrieval-augmented generation, and cloud-based MLOps, while continuing to ship independent projects — from AI chatbots and object-detection systems to assistive hardware — using free-tier cloud infrastructure. He is fluent in English and Hindi and is actively seeking AI/ML engineering or data analyst opportunities where he can apply this combined technical and analytical skill set.

PERSONAL INFORMATION

Location: Prayagraj, Uttar Pradesh, India • **Phone:** +91 9889344818 • **Email:** kushagra.gupta.ald@gmail.com
Languages: English (Fluent), Hindi (Native)

EDUCATION

B.Tech, Computer Science & Engineering (Data Science)

2024 – 2028 (Expected)

Babu Banarasi Das Institute of Technology and Management (BBDITM), Lucknow, India

CGPA: 7.0 (Semester 1) | 7.4 (Semester 2)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Statistics & Probability, Linear Algebra, Computer Networks.

PROFESSIONAL EXPERIENCE — JOB SIMULATIONS

Software Engineering Job Simulation — Hewlett Packard Enterprise (Forage)

August 2026

- Completed a simulation of HPE's Software Engineering team, including proposing and building a RESTful web service from scratch.
- Implemented data-upload functionality and wrote unit tests to validate service behavior.

Quantitative Research Job Simulation — JPMorgan Chase & Co. (Forage)

June 2026

- Completed a simulation of JPMorgan Chase's Quantitative Research division, applying structured analytical and financial-reasoning methods to real-world style datasets.
- Practiced translating business questions into quantitative research tasks consistent with buy-side/sell-side research workflows.

GenAI Powered Data Analytics Job Simulation — Tata (Forage)

April 2026

- Applied generative AI techniques to solve real-world data analytics problems, including prompt-driven insight generation and automated reporting.
- Explored how large language models can accelerate exploratory data analysis and business communication.

Cybersecurity Analyst Job Simulation — Tata (Forage)

April 2026

- Simulated the responsibilities of a cybersecurity analyst, including threat identification, log analysis, and incident response planning.
- Practiced structured risk assessment and communication of security findings to non-technical stakeholders.

Data Analytics Job Simulation — Quantum (Forage)

February 2026

- Executed an end-to-end data analytics workflow — extraction, cleaning, transformation, and exploratory analysis — on 250,000+ retail transaction records.
- Conducted customer segmentation and purchasing-behavior analysis to uncover high-value cohorts and seasonal buying trends.
- Delivered a comprehensive analytical report with actionable, commercially focused recommendations for data-driven business strategy.

Data Analytics Job Simulation — Deloitte Australia (Forage)

November 2025

- Designed an interactive Tableau dashboard with multiple views, filters, and KPI cards for business performance visualization.
- Performed classification, trend analysis, and anomaly detection using Excel (VLOOKUP, pivot tables, conditional formatting).
- Synthesized insights into a strategic presentation deck supporting executive-level decision-making.

PROJECTS

Bharat AI Assistant V1

Stack: React, Next.js, OpenRouter API, Vercel, JavaScript

A lightweight, full-stack AI chatbot platform enabling users to interact with advanced large language models through a responsive web interface, demonstrating rapid deployment of AI assistants using scalable, free-tier cloud infrastructure.

- Integrated multiple LLMs via the OpenRouter API behind a secure serverless layer, keeping API keys off the client.
- Built a real-time, responsive chat interface accessible without complex login systems.
- Deployed on Vercel with continuous deployment from GitHub for zero-downtime updates.

Cybersecurity: Suspicious Web Threat Analysis

Stack: Python, Scikit-learn, FastAPI, Streamlit, Docker

A production-ready pipeline for detecting and analyzing suspicious web traffic collected from AWS CloudWatch, combining unsupervised and supervised machine learning with a deployable prediction API and dashboard.

- Trained an IsolationForest model for unsupervised anomaly detection and a RandomForest classifier for supervised threat classification, with a shared StandardScaler preprocessing pipeline.
- Built a FastAPI prediction service and a Streamlit dashboard for interactive traffic analysis.
- Documented the full model card (input schema, performance notes, limitations) and containerized the service with Docker for reproducible deployment.

J.A.R.V.I.S. Agent System

Stack: Groq (Llama 3.1) / OpenRouter API, Web Speech API, TensorFlow BlazeFace, Vercel Serverless Functions

A cinematic AI assistant combining voice interaction, real-time face tracking, live weather, and a HUD-style interface, deployed as a secure serverless application.

- Implemented voice input/output using the browser's Web Speech Recognition and Synthesis APIs at zero cost.
- Added real-time face tracking with TensorFlow.js BlazeFace loaded from CDN.
- Secured the LLM API key inside a Vercel serverless function so it is never exposed to the browser.

SENTIENT_LENS

Stack: TensorFlow.js (COCO-SSD), IndexedDB, Web Speech API, GitHub Pages

A fully serverless, browser-based object-intelligence application offering real-time object detection, an AI voice personality, and persistent local storage — with no backend, no API keys, and zero hosting cost.

- Ran real-time human/animal/vehicle detection client-side using TensorFlow.js COCO-SSD.
- Built a 50+ response AI personality engine with contextual voice reactions to detections.
- Persisted detection history, pet captures, and vehicle records in IndexedDB, keeping all user data on-device for privacy.

Supermart Grocery Sales — Retail Analytics & ML

Stack: Python, Pandas, Scikit-learn, Jupyter

An end-to-end analytics repository for a grocery retail sales dataset, covering exploratory data analysis, feature engineering, and comparative model evaluation.

- Performed exploratory data analysis to identify sales trends, seasonality, and category-level performance.
- Engineered features and trained/compared multiple models to evaluate predictive performance on sales data.

Neon Air Draw Ultra PRO

Stack: MediaPipe Hands, Canvas API, JavaScript (ES6)

A gesture-controlled 3D drawing application that transforms a webcam feed into a light canvas, fully operable without a mouse or keyboard via real-time hand-landmark tracking.

- Implemented millimetric hand tracking with Google's MediaPipe Hands for gesture-based drawing, erasing, and mode switching.
- Built a custom particle physics engine for a gesture-activated rainbow trail effect with real-time HSL color cycling.
- Added Z-depth sensitivity so stroke width responds to how far the hand is from the camera, plus a persistent snapshot gallery with one-click PNG export.
- Live demo: kushagra486.github.io/neon-air-paint-

ResumeAI

Stack: Claude API, GitHub API, JavaScript, HTML/CSS

An AI-powered resume generation tool that combines the Claude API with live GitHub API data to help users build and refine professional resumes.

- Integrated the Claude API to generate and refine resume content from user-provided profile and project data.
- Pulled live repository data via the GitHub API to auto-populate project sections.
- Deployed as a live tool at kushagra486.github.io/resumeai, linked directly from the main developer portfolio.

Blind Assist Bot

Stack: Arduino Uno, C++, Ultrasonic Sensors

An embedded assistive-hardware project designed to help visually impaired users navigate their surroundings safely using real-time obstacle detection.

- Programmed an Arduino Uno in C++ to read ultrasonic distance sensors and trigger proximity alerts in real time.
- Designed the alert logic to escalate feedback intensity as obstacles get closer, prioritizing low-latency response.

Smart Budget Assistant

Stack: Python, Pandas, NumPy, Flask

An AI-powered personal finance tool that helps users track spending, categorize transactions, and understand their budget at a glance.

- Built data-processing pipelines with Pandas and NumPy to clean and categorize transaction data.
- Exposed budgeting insights through a lightweight Flask web application.

bharat-ai-demo

Stack: Python, OpenCV, Computer Vision Models

A computer-vision demo suite combining several perception capabilities into one repository for rapid experimentation.

- Implemented facial recognition, voice assistance, and object/emotion recognition modules.
- Added aerial hand-gesture recognition for touchless control scenarios.

kushagra486.github.io — Developer Portfolio

Stack: HTML5, CSS3, JavaScript, GitHub API, GitHub Pages

A self-updating, cyberpunk-themed developer portfolio that fetches all public GitHub repositories in real time, so new projects appear automatically with zero manual maintenance.

- Built a zero-backend, single-file portfolio that calls the GitHub API client-side with pagination to list every public repo, live stats, and language filters.
- Added a Certifications Gallery with tab filters (All/AWS/Forage/Other) and a lightbox viewer for certificate images.
- Implemented a canvas-based particle network background, animated skill bars, and a terminal-style interactive 'About' panel.

TECHNICAL SKILLS

Languages

Python, Java, C, JavaScript, TypeScript, HTML5, CSS3, SQL

AI / Machine Learning

PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, Claude API, OpenAI API, Groq API, Prompt Engineering

Web & Application Frameworks

React, Next.js, Flask, FastAPI, Node.js, Tailwind CSS, Bootstrap

Databases

MySQL, PostgreSQL, MongoDB, Firebase, Supabase

Data & Analytics

Pandas, NumPy, Power BI, Tableau, Excel, Matplotlib, Seaborn, Jupyter

DevOps & Cloud

Git, GitHub Actions, Docker, Vercel, GitHub Pages, Hugging Face Spaces, Netlify

Embedded Systems & Hardware

Arduino Uno, C++, Ultrasonic Sensors

CERTIFICATIONS

Introduction to Generative AI: Art of the Possible

June 2026

AWS Training & Certification

Foundational overview of generative AI concepts, use cases, and responsible-AI considerations, delivered by AWS Training & Certification.

AWS Machine Learning Engineer Associate — Curriculum Overview

June 2026

AWS Training & Certification

Structured overview of the skills and knowledge domains covered by the AWS ML Engineer Associate certification path, including model building, deployment, and MLOps.

Foundations of Prompt Engineering

June 2026

AWS Training & Certification

Core techniques for designing effective prompts for large language models, including prompt structuring, few-shot examples, and iterative refinement.

Quantitative Research Job Simulation

June 9, 2026

JPMorgan Chase & Co. — Forage

Completed practical tasks in investigating and analyzing price data, pricing a commodity storage contract, credit risk analysis, and bucketing FICO scores.

Software Engineering Job Simulation

August 13, 2026

Hewlett Packard Enterprise (HPE) — Forage

Completed practical tasks in creating a proposal for a RESTful web service, building a basic RESTful web service, implementing data upload functionality, and writing unit tests.

GenAI Powered Data Analytics Job Simulation

April 13, 2026

Tata — Forage

Completed practical tasks in exploratory data analysis and risk profiling, predicting delinquency with AI, business reporting and data storytelling, and implementing an AI-driven collections strategy.

Cybersecurity Analyst Job Simulation

April 13, 2026

Tata — Forage

Completed practical tasks in identity and access management (IAM) fundamentals, IAM strategy assessment, crafting custom IAM solutions, and platform integration.

Generative AI Fundamentals

2026

Alison

Self-paced course covering the fundamentals of generative AI models, applications, and limitations.

Data Analytics Job Simulation

February 1, 2026

Quantum — Forage

Completed practical tasks in data preparation and customer analytics, experimentation and uplift testing, and analytics and commercial application.

Data Analytics Job Simulation

November 12, 2025

Deloitte Australia — Forage

Completed practical tasks in data analysis and forensic technology.

IT Support Certificate

2024

Data Expert Technical Institution

Foundational certificate covering IT support fundamentals, troubleshooting, and technical customer service.

KEY ACHIEVEMENTS

- Completed 6 industry job simulations across finance (JPMorgan Chase), enterprise software engineering (HPE), technology consulting (Tata, Deloitte Australia), and retail analytics (Quantum).
- Earned 11 professional certifications, including 3 from AWS Training & Certification, within a single academic year.
- Built and deployed 11 production AI/ML, computer-vision, embedded-systems, and full-stack applications, all live and publicly accessible at zero or near-zero hosting cost.
- Designed a self-updating developer portfolio that automatically surfaces new GitHub projects via the GitHub API with zero manual maintenance.
- Delivered a retail analytics report on 250,000+ transaction records with actionable, commercially focused recommendations.
- Built a production-style machine learning pipeline combining unsupervised and supervised models for cybersecurity threat detection.

AREAS OF INTEREST

Industry Areas: FinTech and quantitative analytics, cybersecurity and threat intelligence, AI-powered developer tooling, assistive technology, open-source software development.

Currently Exploring: Retrieval-Augmented Generation (RAG) pipelines, multi-agent AI architectures, AWS SageMaker for MLOps, advanced SQL analytics, and React Native for mobile AI applications.

REFERENCES

Available upon request. Academic, professional, and project-based references can be provided from faculty supervisors and Forage simulation coordinators.